

Article

DK-PRACTICE: An Intelligent Platform for Knowledge Tracing and Educational Content Recommendation: A Case Study in Higher Education

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Abstract

This paper introduces DK-PRACTICE, an intelligent educational platform that combines Knowledge Tracing (KT) and recommendation systems to support personalized learning in higher education. The platform utilizes a novel Paired Bipolar Bag-of-Words (PB-BoW) model to assess students' knowledge states, forecast performance, and offer targeted recommendations. To test its effectiveness in real-world settings, DK-PRACTICE was implemented in the "Computer Organization and Architecture" undergraduate course, involving 138 students in Pre-Test and 106 in Post-Test. Empirical analysis of benchmark datasets and a newly created course dataset showed that the PB-BoW model outperformed an RNN-based KT model in predictive accuracy. Student surveys indicated high levels of satisfaction with usability, relevance of recommendations, and overall learning support, with most participants expressing willingness to reuse the platform in other courses. These results demonstrate the potential of DK-PRACTICE as a scalable and adaptable tool for improving personalized learning and bridging the gap between AI-driven KT research and classroom implementation.

Keywords: knowledge tracing; educational recommendations; machine learning; intelligent educational platform; education



Academic Editor: Aneta
Poniszewska-Maranda

Received: 24 December 2025

Revised: 8 February 2026

Accepted: 13 February 2026

Published: 15 February 2026

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1. Introduction

Addressing the educational challenges, such as accurately assessing student knowledge, predicting performance, and providing tailored educational content, is essential for effective personalized learning. This alignment is particularly evident in Competence-Based Learning (CBL) frameworks, where the focus shifts from traditional time-bound instruction to the mastery of specific skills and competencies. This study introduces DK-PRACTICE, a Knowledge Tracing (KT)-based platform developed by utilizing Machine Learning (ML) methods to recommend the appropriate educational content to cover students' knowledge gaps.

Machine learning, as a cornerstone for Artificial Intelligence (AI), is rapidly reshaping educational practices by enabling personalized, adaptive, and interactive learning experiences. In online educational systems, tools based on machine learning methods and, by

extension, on artificial intelligence have shown particular progress. As noted in [1], the integration of tools such as PhET [2], Labster [3], ChatGPT [4], Squirrel AI [5], and IBM Watson [6] in the educational process led to a 28% improvement in student performance in many institutions. The study also found increased student motivation, conceptual clarity, and satisfaction, particularly when using simulation-based environments. These results demonstrate that AI can enhance student performance. Learning effectiveness is enhanced, especially when implemented through tools designed to personalize teaching and respond to students' individual needs.

Knowledge Tracing (KT), a modeling technique that assesses students' knowledge evaluation in educational contexts, like skills and concepts, is positioned at the core of many educational intelligent systems. KT allows systems to monitor students' progress and suggest tailored content based on predicted knowledge gaps. Early KT models, such as Bayesian Knowledge Tracing (BKT) [7] and Item Response Theory (IRT) [8], laid the groundwork for individualized teaching with a possible assessment of learning skills performance. More recent developments, including Deep Knowledge Tracing (DKT) [9] and Graph-based Knowledge Tracing (GIKT) [10], use repetitive neural networks and neural graph networks to better model learning time patterns and relational dependencies between skills and questions. By granularly mapping student performance to individual concepts, these KT models provide the necessary diagnostic data to operationalize CBL, allowing for a shift toward proficiency-based progression [11].

Despite these technical advances, many AI-powered KT systems remain in a laboratory testing stage and have not been tested in real educational environments. Although platforms such as Knewton Alta [12], Carnegie Learning's MATHia [13], and Squirrel AI [5] have demonstrated efficacy in controlled or proprietary environments, there is a lack of research examining how KT systems function when deployed in real-world classrooms with diverse student populations. In addition, advanced models such as AdvKT [14], TrueLearn [15], TutorLLM [16], and ACKT [17] have focused on simulation fidelity or probabilistic feedback, but most lack classroom validation or accessible application frameworks for educators.

To address this gap, our study implements a case study using the DK-PRACTICE platform and evaluates the results of applying this knowledge assessment platform in a real educational environment. DK-PRACTICE monitors student knowledge levels and, taking into account student knowledge representation and a recommendation algorithm, recommends personalized educational content. To ensure pedagogical consistency and scalability, the platform's data model is aligned with the standardized Body of Knowledge (BoK) and Curriculum Body of Knowledge (CBok) frameworks. By mapping learning concepts to formal Knowledge Units (KUs), DK-PRACTICE facilitates a systematic assessment of student proficiency that is consistent with international academic standards [18,19]. In our case study, DK-PRACTICE was tested with 98 2nd-year university students in the "Computer Organization and Architecture" course. This approach facilitates a CBL environment by ensuring that students address fundamental architectural concepts through targeted remediation before moving to more advanced topics.

The present study investigates the following research questions:

- RQ1: How can a fast and efficient knowledge tracing model, based on BoW-inspired representation of student knowledge, be developed to effectively identify learning gaps through **Pre-** and **Post-test** assessments?
- RQ2: What is the impact of personalized content recommendations generated by the DK-PRACTICE platform, and how do they improve learning outcomes?
- RQ3: How could the DK-PRACTICE platform enhance the educational process, and how could its use be extended to other educational topics?

Key contributions of this work include: (1) the introduction of a new lightweight ML method for knowledge tracing based on a paired bipolar bag-of-words representation of students' knowledge, (2) the design and actual implementation of the DK-PRACTICE platform as a case study in a real-world learning environment, (3) an empirical assessment of the impact of the test operation, and (4) a comprehensive comparison of DK-PRACTICE with existing AI-based KT systems, demonstrating its potential as a scalable and effective tool for adaptive learning.

Section 2 presents the current scientific picture of the evolution of knowledge tracing and educational recommendation models. The following Section 3 describes the technologies used for the development of the DK-PRACTICE educational platform and the realization of this study. In Section 4, the platform and its functions are presented. Section 5 describes the case study, with the analysis of the results commented on in Section 6. Section 7 concludes the study and presents the future extensions and improvements of the DK-PRACTICE platform.

2. Related Works

Artificial Intelligence (AI) is emerging as a pillar of transformation in education, offering new ways to personalize learning and improve learning outcomes. Examples include AI-powered platforms such as PhET [2] and Labster [3], providing learning experiences based on simulating lessons such as physics or mathematics, while interlocutors such as ChatGPT 4 [4] support on-demand clarification and problem solving. The integration of AI tools into structured 60 min virtual physics lessons led to a 28% increase in student performance, along with improvements in engagement, conceptual clarity, and motivation [1], validating the potential of AI-enhanced systems to address pedagogical challenges, including limited interactivity in the classroom.

The common benchmark in many of these AI systems is Knowledge Tracing (KT)—the task of assessing a student's evolving knowledge of skills or concepts over time based on the results of interaction with educational platforms and observed behavior [7]. Early KT models, such as Bayesian Knowledge Tracing (BKT) [20] and the Item Response Theory (IRT) [8], applied probabilistic frameworks to infer students' knowledge states. The evolution in KT's field of research was brought about by the introduction of deep learning, specifically the Deep Knowledge Tracing (DKT) model by [9], which utilized recurrent neural networks (RNNs) to capture time dependencies and non-linear learning patterns. Subsequent improvements were presented in the Dynamic Key-Value Memory Networks (DKVMN) [21], which separated the representation of concepts from the mastery level to enhance interpretability, and Self-Attentive Knowledge Tracing (SAKT) [22], which utilizes transformer-based attention mechanisms to prioritize critical learning events. These innovations allowed for more accurate prediction and identification of knowledge gaps. In addition, the Bag-of-Words (BoW) representation method originating from natural language processing can also be used as an approach for knowledge vector representation in machine learning. The BoW method simplifies unstructured data into fixed-length feature vectors by tracking how often individual tokens appear, setting aside token order [23].

Further extensions to KT have improved both accuracy and personalization. Exercise-Enhanced Sequential Modeling (EKT) introduced exercise-level features such as difficulty and subject to improve the accuracy of recommendations [24], while the Bidirectional GRU Knowledge Tracing (Bi-GRU) model [25] introduced forward and regressive diagnosis for both predicting performance and estimating students' knowledge states from previous interactions, combined with a feedback network to predict the answer correctness in upcoming questions.

Online educational systems, along with practical applications on platforms such as ALEKS [26], Knewton [12], and Duolingo [27], highlight the role of KT in providing personalized learning experiences in real-time.

In recent research, ref. [28] introduced two advanced KT models—Knowledge Structure-aware GraphAttention Network (KSGAN) and Knowledge Concept-based Memory Network (KCMN)—that model the complex relationships between exercises and knowledge concepts. By combining graph attention networks and memory mechanisms, their approach enhances interpretability and detail in predicting student performance. In simulation research, the authors of [29] addressed the problem of bias accumulation in KT-based simulators by proposing DSIm. This conditional diffusion model best approximates the actual behavior of the students. This innovation improves the quality of synthetic data used to train recommendation engines, overcoming the limitations of traditional KT models in production tasks. On the student engagement side, ref. [30] conducted a large-scale randomized trial evaluating Squirrel AI, showing significant improvements in student motivation and engagement. The study highlighted that AI-based adaptive systems not only enhance cognitive performance but also support non-cognitive factors vital for learning success.

Within this landscape of increasingly sophisticated models, most remain either research prototypes or validated in synthetic or comparative datasets. This study introduces DK-PRACTICE, a KT-based platform specifically designed for classroom deployment in the real world. DK-PRACTICE assesses students' knowledge levels and recommends personalized pre-test diagnostic-based content. DK-PRACTICE offers the ability to monitor changes in students' knowledge state across learning objects or concepts, individually or in groups. Although many advanced models such as AdvKT [14], GIKT [10], or TutorLLM [16], contribute valuable architectural innovations such as multi-step simulation, graph-based integrations, and language model integration, they often lack deployment scenarios that evaluate educational impact, student feedback, and scalability under real-world constraints and have not yet been tested in classroom environments. DK-PRACTICE addresses these gaps by prioritizing functionality aligned with classroom practices, including diagnostic assessments, actionable feedback, and content alignment.

In summary, the literature illustrates a rich and rapidly evolving area of research in knowledge discovery and AI-assisted individualized training with models that have laid the technical foundations, such as DKT, DKVMN, SAKT, etc. However, few systems effectively bridge the gap between research and practical application. DK-PRACTICE contributes to this space by combining a robust machine learning model with real educational development, proving that adaptive KT platforms can be not only technically innovative but also pedagogically effective.

3. Background Technologies

The DK-PRACTICE online education platform was built using machine learning methods. Machine Learning (ML), a prominent subfield of Artificial Intelligence (AI), has been extensively applied in educational contexts. As previously discussed, ML algorithms have played a substantial role in the analysis of student behavior [31], the prediction of academic performance [32], and the provision of personalized learning recommendations [33].

From a theoretical point of view, neural networks constitute a core framework for modeling complex nonlinear mappings in high-dimensional spaces [34,35]. Neural networks have emerged as a foundational paradigm in the advancement of machine learning techniques [36]. Empirical evidence demonstrates that neural networks, particularly multilayer feedforward architectures, consistently improve performance in domains requiring the extraction of complex feature hierarchies from raw, unstructured data [37]. Their capacity for end-to-end learning demonstrates the ability to autonomously identify and encode

hierarchical feature representations from unstructured input data, thereby enabling improved performance across a wide range of complex pattern recognition tasks [38]. These technological advances form the foundation for modern adaptive learning platforms such as DK-PRACTICE.

Educational recommendation systems are smart technologies designed to personalize learning by suggesting relevant content, activities, or learning paths based on a student's level of knowledge and performance history. Their primary role is to support adaptive learning by filling knowledge gaps with educational proposals tailored to individual needs [39]. Unlike traditional static curricula, these systems dynamically adapt in real time, helping students engage with the right content at the right time to maximize efficiency and learning motivation. Beyond their established contribution to academic outcomes [40], modern education systems are evolving through data-driven modeling. Techniques like machine learning and knowledge tracing (KT) are now central to this evolution, providing the tools needed to analyze and improve student performance. In classroom settings, recommendation systems also help teachers by automating feedback, identifying student weaknesses early, and facilitating data-driven teaching.

As education becomes increasingly digitalized, these systems play a crucial role in scaling up personalized learning experiences. Educational recommendation systems can be broadly categorized into various types based on the algorithms and data sources they use. The most common approaches include content-based filtering, which suggests elements similar to what a student has previously interacted with, and collaborative filtering, which leverages patterns of behavior among like-minded users [41,42]. In the field of education, more advanced categories of recommendation systems have emerged. Knowledge-based systems use predefined concept hierarchies or knowledge graphs to align learning content with curriculum structures [43]. Context-aware systems adjust recommendations based on external variables such as time, location, student goals, or engagement history [44,45], and hybrid systems combine multiple techniques to improve accuracy and personalization [46]. These advanced systems are often augmented with KT algorithms to monitor learning progress and ensure that quality aligns with students' evolving knowledge.

4. The DK-PRACTICE Platform

In this study, we introduce DK-PRACTICE (Figure 1), an intelligent online platform designed to conduct adaptive knowledge assessments through a question–answer method, wherein each question is dynamically tailored based on students' prior responses. The intelligent DK-PRACTICE functions as an integrated knowledge tracing and recommendation system and aims to

- Accurately estimate students' current knowledge states;
- Predict their future performance on subsequent personalized questions;
- Recommend targeted instructional content to address specific learning objectives, concepts, and skills where deficiencies have been identified;
- Support users' roles.

The knowledge tracing component, called PB-BoW (Paired Bipolar Bag of Words), is inspired by the Bag of Words approach [47] and is implemented using a feedforward neural network, which allows modeling of the dynamic and sequential nature of interactions between students and the platform. In NLP the BoW representation works by treating words as separate elements in a vocabulary which are either present or not present in a sentence. Therefore, one bit is enough to represent each word in the BoW vector. Consequently, BoW can be efficiently used to derive informational value from the co-occurrence of words in the sentences. In our context, however, questions (words) not only appear in the exams (sentences), but students can also respond to them with two possible outcomes: wrong

or correct. For this reason, we propose to extend the BoW representation using 2 bits per question, one for the correct answer and one for the wrong answer. Our approach facilitates robust and accurate predictions of students' evolving knowledge states. Educational recommendations are generated based on the predicted probability of correctness for the next unattempted question, thereby guiding the selection of instructional resources aimed at remediating identified knowledge gaps. DK-PRACTICE supports three distinct user roles: "administrator", "student", and "tutor". Students engage with adaptive assessments, tutors construct and manage test content, and administrators oversee system operations. A comprehensive description of the platform's architecture, functionalities, and implementation details is provided in the following sections.

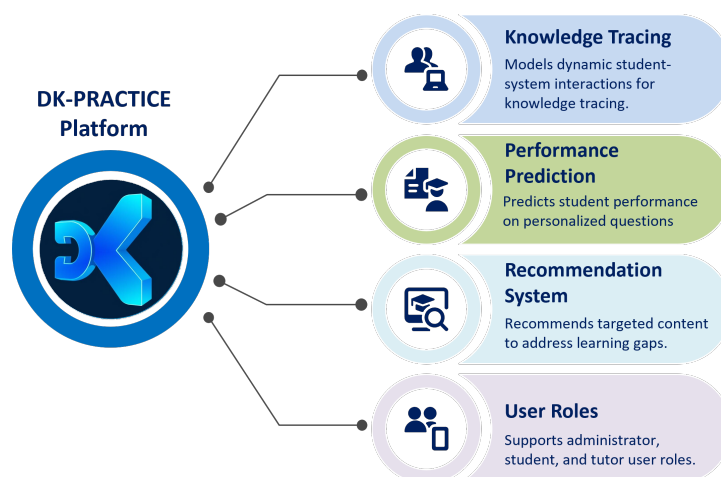


Figure 1. The DK-PRACTICE platform functionalities.

4.1. The DK-PRACTICE Platform Architecture and Functionalities

The architecture of the DK-PRACTICE platform (Figure 2) includes two main components based on machine learning methods:

- The knowledge tracing component to predict students' performance and to estimate students' knowledge state, and
- The recommendation component to produce educational content recommendations.

The architectural structure of the DK-PRACTICE platform not only enhances the fidelity of the knowledge state assessment but also makes the model suitable for auxiliary tasks, such as personalized recommendations. Both knowledge tracing and recommendation components constitute core parts of the DK-PRACTICE platform, and their implementation resides within the back-end architecture.

4.2. Data Model

Data is one of the most fundamental components of the DK-PRACTICE platform. The development of the DK-PRACTICE data model followed a systematic mapping process to ensure alignment with the Curriculum Body of Knowledge (CBoK). Each question was first categorized based on the course's learning objectives and then cross-referenced with the standardized ACM/IEEE Computer Science Body of Knowledge (BoK) [18,19]. Specifically, questions were mapped to distinct Knowledge Units (KU) and Knowledge Areas (KA), as described below. This hierarchical structure (Concept→KU→KA) allows the recommendation engine to not only address immediate knowledge gaps but also to provide a broader assessment of the student's proficiency within the standardized domain-dependent framework required by the curriculum. Consequently, the system's diagnostic capabilities are grounded in a globally recognized educational standard, facilitating more robust and interoperable student performance analytics.

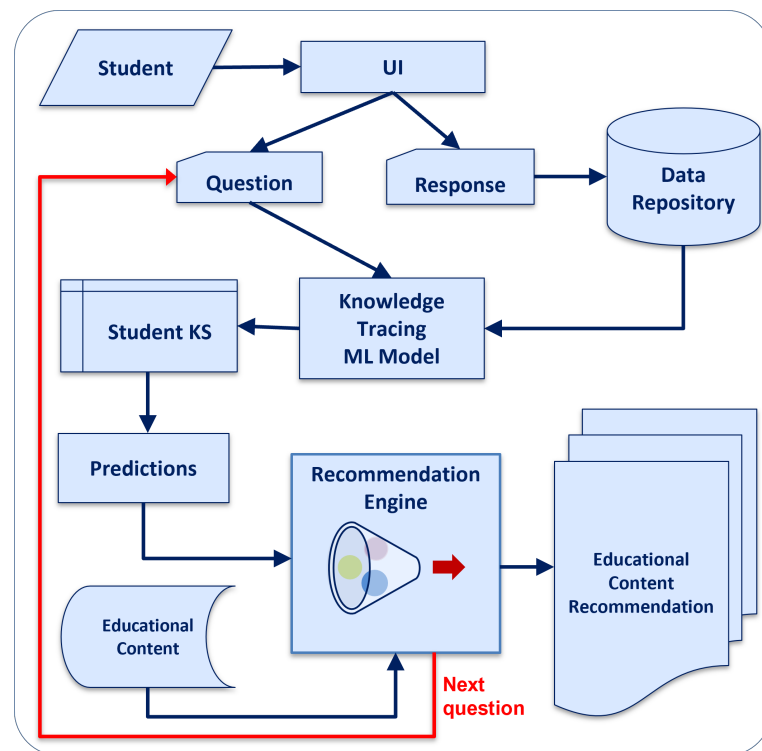


Figure 2. The DK-PRACTICE platform architecture.

The data model (Figure 3) represents the fundamental infrastructure of the platform, enabling the machine learning algorithms to process and interpret student knowledge states effectively. The database schema is organized around three primary entities: “Concepts”, “Questions”, and “Exams”. Within the “Questions” entity, the “Chapter” attribute denotes the specific segment of the educational material associated with the question.

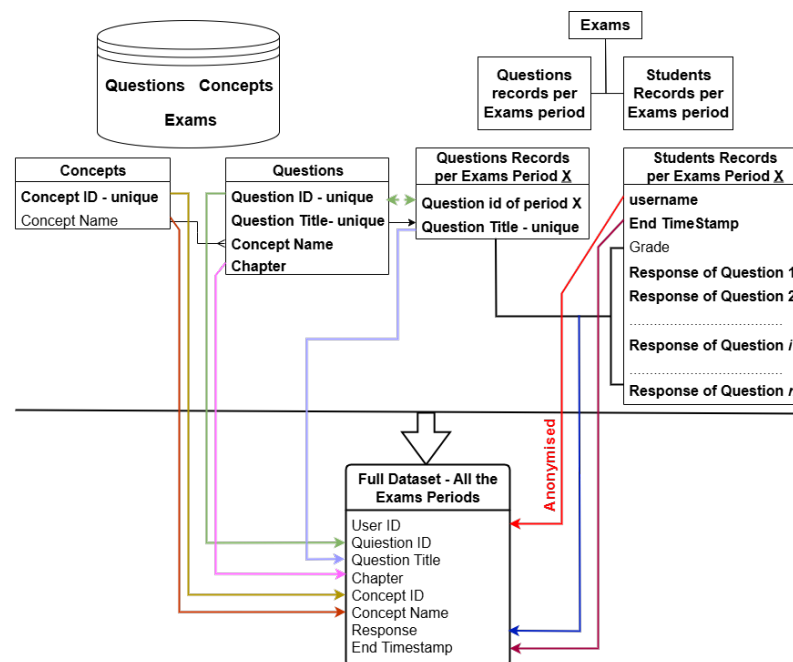


Figure 3. The data model.

The compiled database consists of the following:

- Examination questions,
- The corresponding underlying concepts,

- The chapters containing the educational content of the course, and
- Students' submitted answers across different examination periods.

The concepts are systematically mapped to both the exam questions and the educational content chapters, thereby establishing a structured linkage between learning materials and the questions. This attribute serves as a key parameter in the recommendation algorithm, enabling the identification and prioritization of relevant educational content when generating personalized recommendation lists.

The dataset is derived from the database and consists of interaction records, specifically question–answer pairs (q, r) , recorded within the “Exams” entity. The responses are binary: $r = 0$ for an incorrect answer and $r = 1$ for a correct answer. All personally identifiable information was removed through an anonymization process to ensure compliance with ethical and privacy standards. The constructed dataset was subsequently utilized to train the machine learning model designed to infer the knowledge tracing component, described below, was conducted by leveraging students' historical interaction data.

4.3. Knowledge Tracing Component

The concept of [25], in terms of handling question–response interactions, inspired the knowledge tracing model that we introduce and utilize in the DK-PRACTICE platform. Unlike previous deep learning knowledge tracing models [1,7,9,21,22], our model explicitly disconnects the current question from the current representation of knowledge. This leads to the production of the knowledge state representation vector in which the student's learning is incorporated, but not including the current time step τ , and is based exclusively on the sequence of previous question–answer pairs (q_k, r_k) , $k = 1, \dots, \tau - 1$.

For our knowledge tracing model we propose an extension of BoW using paired bipolar representation of answered questions (Figure 4). The representation vector $\mathbf{x}$ has a length of $2 \cdot N_q$, where N_q is the total number of questions. All elements of $\mathbf{x}$ are zero except that for each question i with an existing response r , the following values are assigned:

- If $r = 0$ (incorrect answer), then $x_{2i-1} = 1$ and $x_{2i} = -1$.
- If $r = 1$, (correct answer) then $x_{2i-1} = -1$ and $x_{2i} = 1$.

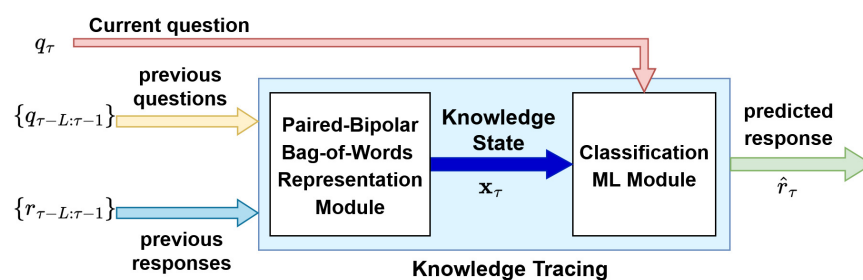


Figure 4. The Paired Bipolar Bag-of-Words (PB-BoW) knowledge tracing model.

The corresponding representation scheme will be called Paired Bipolar Bag-of-Words (PB-BoW).

A total of $M = 200$ samples are generated for each student through the following procedure:

- A subset of 10 questions is randomly selected to form the **query set** for training. The remaining questions are designated as the **evaluation set**.
- The PB-BoW vector representation, $\mathbf{x}$, is generated for the query set. This vector has a length of $2 \times N_q$ and serves as the input to the classification module.
- The target vector, $\mathbf{t}$, for the classification module is constructed using the responses from the evaluation set. All elements of $\mathbf{t}$ are equal to -1 except for each evaluation question q_j with response r_j , where we set $t_j = r_j$. Therefore, unanswered questions have target value $t = -1$, allowing us to exclude them when calculating the loss of the classifier.

The Classification Module

The task of the classification module is for every student to predict the responses to all questions, given a history of 10 prior interactions (question/response pairs). The classifier is a feedforward neural network with the following layer specifications:

- **Input Layer:** $2 \times N_q$ neurons
- **Hidden Layers:** Three fully-connected hidden layers.
 - Layer 1: 15 neurons.
 - Layer 2: 10 neurons.
 - Layer 3: 5 neurons.
- **Output Layer:** N_q neurons. Each neuron corresponds to a question, and its value, between 0 and 1, predicts the probability of a correct response.

The input to the classifier is the PB-BoW vector representation $\mathbf{x}$ of the student's interactions in the query questions. The target of the classifier is the vector $\mathbf{t}$ corresponding to the evaluation questions.

The set of students is split into training and test subsets. For each training student s , we generate the corresponding dataset D_s^{train} according to Algorithm 1. The overall training set results from the concatenation of all D_s^{train} datasets. In a similar fashion, we generate the test set from the datasets D_s^{test} of the test students.

Algorithm 1 Generate Training Samples for PB-BoW KT

```

1: procedure GENERATESAMPLES( $X$ )
2:   Input: Questions and responses data  $X = \{(q_1, r_1), (q_2, r_2), \dots, (q_{N_{qs}}, r_{N_{qs}})\}$  for student  $s$ . The
     values  $q_i$  are the indices of the questions answered by the student and  $r_i$  are the corresponding
     responses (0 or 1);  $N_{qs}$  is the number of questions answered by this student.
3:   Output: Dataset  $D_s = \{(\mathbf{x}_i, \mathbf{t}_i)\}_{i=1}^{200}$ 
4:   Let  $N_q$  be the total number of unique questions in the question bank.
5:   Set  $D_s \leftarrow \emptyset$ 
6:   for  $i = 1$  to 200 do
7:     Randomly select a subset of 10 questions to form the query set  $Q \subset X$ .
8:     The remaining questions form the evaluation set  $E = X \setminus Q$ .
9:     Initialize input vector  $\mathbf{x}$  of size  $2 \times N_q$  with zeros.
10:    for each question-response pair  $(q_j, r_j) \in Q$  do
11:      Generate Paired-Bipolar representation:
12:      if  $r_j = 0$  then                                     ▷ Incorrect response
13:         $x[2q_j] \leftarrow 1$ 
14:         $x[2q_j + 1] \leftarrow -1$ 
15:      else                                                 ▷ Correct response
16:         $x[2q_j] \leftarrow -1$ 
17:         $x[2q_j + 1] \leftarrow 1$ 
18:      end if
19:    end for
20:    Initialize target vector  $\mathbf{t}$  of size  $N_q$  with  $-1$ .
21:    for each question-response pair  $(q_k, r_k) \in E$  do
22:       $t[q_k] \leftarrow r_k$ 
23:    end for
24:    Add the pair  $(\mathbf{x}, \mathbf{t})$  to  $D_s$ .
25:  end for
26:  return  $D$ 
27: end procedure

```

The classifier loss L , is the mean squared error computed only on the questions with a valid target value, i.e., with $t_i \neq -1$:

$$L = \frac{1}{N} \sum_{i=1}^{N_q} (t_i - \hat{r}_i)^2 I(t_i \neq -1)$$

where $\hat{r}$ is the classifier output, $I(\cdot)$ is the indicator function ($I(cond) = 1$ if $cond = \text{true}$, and 0 otherwise), and $N = \sum_i I(t_i \neq -1)$.

4.4. Recommendation Component

The platform implements two types of recommendation mechanisms: (a) *next-question recommendation* during a knowledge assessment and (b) *post-assessment recommendation* of educational materials.

4.4.1. Next Question Recommendation

Let the query question set for a student be denoted by

$$Q = \{q_1, q_2, \dots, q_L\}.$$

where $L = 10$. Questions are asked sequentially in an order that depends on the students' previous answers. Therefore, questions are not asked in the same order for every student. The function $m(\tau)$ returns the index of the question to be asked at time step τ . Due to lack of prior information, in the beginning of the sequence, all participants start with the same initial question $q_{m(1)}$, referred to as the *cold-start question*.

Subsequent questions are personalized for each student and adapted according to their prior responses. The recommendation model utilizes the estimated knowledge state vector of the student to compute the probability of a correct response for each remaining question in the question bank. At time τ , the question set is split into two subsets:

1. the set of *Potential questions*, Q_{pot}^τ , which remain eligible to be asked as the next question;
2. the set of *Answered questions* Q_{ans}^τ that are ineligible to be asked again.

Initially, all questions, except for the *cold-start question*, are classified as *Potential* while the cold start question is classified as *Answered*.

At each time step $\tau \geq 2$:

1. The asked question q_τ is moved from $Q_{\text{pot}}^{\tau-1}$ to Q_{ans}^τ :

$$Q_{\text{ans}}^\tau = Q_{\text{ans}}^{\tau-1} \cup \{q_\tau\}, \quad Q_{\text{pot}}^\tau = Q_{\text{pot}}^{\tau-1} \setminus \{q_\tau\}.$$

2. For each $q \in Q_{\text{pot}}^\tau$, the recommendation model computes the predicted probability of a correct response based on the student's estimated knowledge state vector $\mathbf{x}_\tau$:

$$p_\tau(q) = \Pr(\hat{r}_\tau(q) = 1 \mid \mathbf{x}_\tau).$$

3. The next question is selected as the one whose probability of correct response is closest to 0.5,

$$q_{\tau+1} = \arg \min_{q \in Q_{\text{pot}}^\tau} |p_\tau(q) - 0.5|,$$

thereby targeting questions for which the model is maximally uncertain.

Within the framework of the DK-PRACTICE platform, we have established the personalized question quota at $L = 10$. This means that we can only use 10 questions and responses to determine the knowledge state of the student. The remaining questions are used for evaluating the performance of our prediction model. This quota constitutes a hyperparameter and is subject to occasional adjustments following preferences, experimental findings, course requirements, etc.

4.4.2. Educational Content Recommendation

Following the completion of the test, and based on the predicted knowledge state of the student derived from the test results, the probability of correctly answering each question in the question bank is recalculated. Each question q is associated with a learning object or concept $c(q)$.

The knowledge level for a given learning concept c is computed as

$$K_c = \frac{1}{|Q_c|} \sum_{q \in Q_c} p_q$$

where Q_c is the set of questions linked to c , and $|Q_c|$ is its cardinality. The value K_c represents the average estimated probability of correct responses for questions concerning learning object c , and is interpreted as the student's mastery level for that object.

The recommendation engine identifies knowledge gaps not as isolated failures, but as deficiencies within the broader BoK framework. By incorporating the BoK and CBoK dimensions, the engine facilitates a systematic approach to addressing curriculum-related gaps, ensuring that the recommended content supports mastery of the standardized competencies required for the "Computer Organization and Architecture" course. Specifically, the engine dynamically selects educational materials corresponding to the learning concepts with the lowest estimated knowledge state (K_c) values; this approach prioritizes the specific knowledge units where the student demonstrates the weakest mastery, thereby enabling a targeted, proficiency-based remediation process.

All this process is supported by a data model which is organized into a database, a training dataset, and a database of educational content.

4.5. Knowledge Tracing Experiments and Results

In this work, we introduce a new real-world dataset named "COaA course", which was collected from nine examination periods in the undergraduate "Computer Organization and Architecture" course at the Department of Information and Electronic Engineering of the International Hellenic University via the Moodle electronic learning management system. These data encompassed both the period of the COVID-19 pandemic, during which courses and assessments were conducted under quarantine restrictions and the subsequent post-pandemic period. The database constructed for the case study adheres to the platform's underlying data model.

The "COaA course" dataset comprises a total of 19,339 interactions, encompassing question-answer pairs generated by 1115 students on 120 multiple-choice questions, 31 concepts, and 20 chapters of educational content. For experimental purposes, the data set was split into training and testing subsets. Consequently, the training set contains 15,555 interactions from 891 unique students, while the testing set consists of 3784 interactions from 224 unique students. Each student contributes a minimum of 11 interactions (i.e., question-answer pairs), thereby ensuring adequate data density for reliable model evaluation.

In addition to the newly constructed "COaA Course" dataset, developed for evaluating the PB-BoW KT model, we employed two widely used ASSISTments benchmark datasets (<https://sites.google.com/site/assistmentsdata/>, accessed on 10 November 2024): "ASSISTment 2009" and "ASSISTment 2017". The "ASSISTment 2009" dataset comprises 101 concepts, with model inputs represented as corresponding questions. It contains data from 4151 students, including 1196 in the training set and 511 in the test set, yielding a total of 274,590 student-question interactions. Similarly, the "ASSISTment 2017" dataset includes 101 concepts with model inputs likewise represented by questions. This dataset encompasses 1707 students (1196 in the training set and 511 in the test set), resulting in

864,703 recorded interactions. Both datasets include multiple student responses to identical questions, as well as records with fewer than 11 question–answer interactions per student.

For the implementation of the PB-BoW KT model, during the data preprocessing procedure, we excluded records containing fewer than 11 interactions and retained only the first response to each repeated question per student. The resulting dataset statistics, including the number of students and interactions in both the training and test splits, are summarized in Table 1.

The experiments were conducted on two knowledge tracing models: the proposed PB-BoW KT model and the KT-BiGRU [25] model.

Table 1. Dataset statistics after preprocessing. For each dataset, the number of students and total interactions are reported, along with their distribution across training and test sets.

Dataset	Questions	Students			Interactions		
		Total	Train	Test	Total	Train	Test
ASSISTment 2009	101	818	556	262	23,722	16,511	7211
ASSISTment 2017	101	1641	1144	497	69,712	47,856	21,856
COaA Course	120	1115	891	224	19,339	15,555	3784

The results reported in the Table 2 demonstrate that the PB-BoW KT model consistently outperforms the KT-BiGRU, SAKT, and DKT models across all three datasets in terms of both AUC and Accuracy. On the “ASSISTment 2009” dataset, the PB-BoW KT model achieves an AUC of 0.765 and an accuracy of 0.760, exceeding the scores of the deep learning-based models. A similar trend is observed on the “ASSISTment 2017” dataset, where the PB-BoW KT model attains an AUC of 0.733 and an accuracy of 0.700, slightly higher than the corresponding values obtained by its competitors. The most pronounced performance gap appears in the “COaA Course” dataset, where PB-BoW KT achieves the highest results overall with an AUC of 0.805 and an accuracy of 0.787, compared to 0.770 for SAKT and 0.756 for DKT.

Table 2. The best results of AUC and accuracy metrics per dataset and per knowledge tracing model.

Dataset	PB-BoW		KT-BiGRU		SAKT		DKT	
	AUC	Acc	AUC	Acc	AUC	Acc	AUC	Acc
ASSISTment 2009	0.765	0.760	0.753	0.713	0.723	0.759	0.723	0.747
ASSISTment 2017	0.733	0.700	0.716	0.686	0.650	0.733	0.644	0.755
COaA Course	0.805	0.787	0.755	0.719	0.770	0.739	0.756	0.738

These findings indicate that the PB-BoW KT model provides robust and superior predictive performance consistently across different datasets, suggesting its suitability for knowledge tracing tasks. It is noteworthy that simple machine learning approaches can outperform deep learning methods while also offering substantially reduced training time and eliminating the need for GPU resources. Specifically, the PB-BoW KT model was trained on a system equipped with an 11th-generation CPU operating at 2.40 GHz and 16 GB of RAM. For all datasets, training was conducted for 20 epochs, with an average runtime of approximately 5 s per epoch, highlighting the computational efficiency of the proposed PB-BoW approach.

4.6. The DK-PRACTICE Platform Implementation

The DK-PRACTICE platform employs a client–server architecture, with the front-end developed in PHP and JavaScript. Data is stored and managed through a MySQL database schema. In the back-end of the platform, a security-enabled Flask (Welcome to Flask-Flask documentation, <https://flask.palletsprojects.com/en/3.0.x/> accessed on

23 December 2025), (Welcome to Flask-Security documentation, <https://flask-security-too.readthedocs.io/en/stable/> accessed on 23 December 2025)-based API (Application Programming Interface) supports bidirectional communication with the machine learning models, providing outputs such as “Next Question” recommendation and the final “Educational Recommendations”, which support the delivery of personalized educational content. In addition, to ensure full functionality, the platform provides user registration and password recovery services in compliance with security policies.

As mentioned above, the DK-PRACTICE platform supports three types of users: “administrator”, “tutor”, and “student”. Thus, users

- with the “administrator” role (Figure 5). Beyond the general oversight of the platform, the administrator is tasked with the: (a) establishment of courses, (b) the registration of tutors, and (c) the allocation of course management responsibilities to the corresponding tutors.

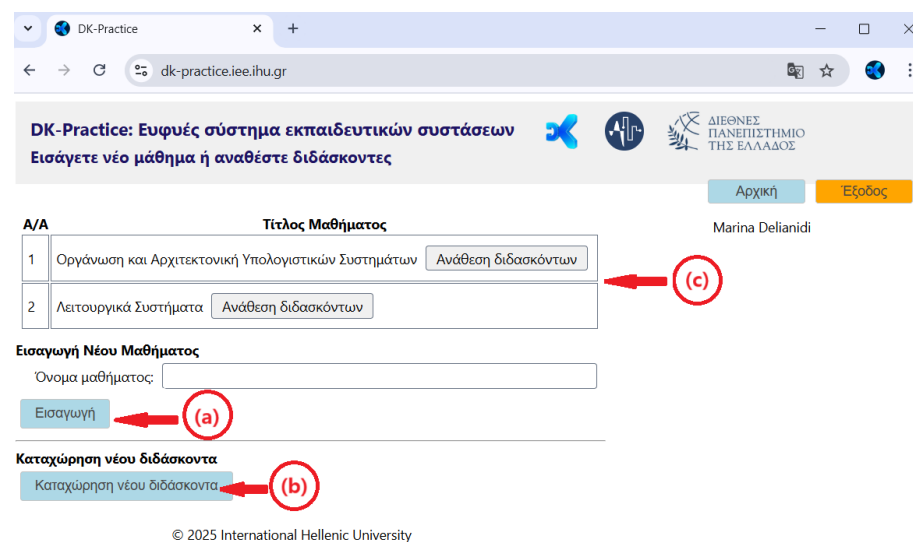


Figure 5. Administrator screen. DK-Practice: The Intelligent Educational Recommendations System. Insert a new course.

- with the “tutor” role have the permissions to create a test (Figure 6) by entering all the relevant information and to set parameters such as starting and ending date and time (Figure 7), and monitor the students’ performance individually and in groups per Test (Figure 8). The ability to monitor test results enables the teacher to have an overview of the class’s knowledge state regarding the specific course, and indeed for each educational object or concept.

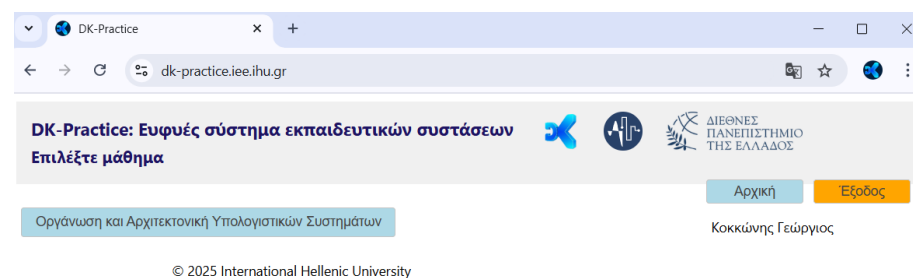


Figure 6. Tutor screen: the allocated courses. DK-Practice: The Intelligent Educational Recommendations System. Choose the course.

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DK-Practice: Ευφυές σύστημα εκπαιδευτικών συστάσεων





ΔΙΕΥΘΥΝΣΗ ΠΑΝΕΠΙΣΤΗΜΙΟ ΤΗΣ ΕΛΛΑΔΟΣ

Αρχική

Έξοδος

Κοκκίνης Γεώργιος

Αρχείο τεστ

2025-05-28

2025-05-19

2025-05-18

2025-05-05

2025-05-04

2025-05-03

2025-05-01

2025-01-11

2025-01-04

Νέα εξέταση

Έναρξη εξέτασης: Set the Start of the Test. Year, Month, Day, Hour, Minutes

Έτος

2025

Μήνας

8

Ημέρα

11

Ώρα

15

Λεπτό

45

Λήξη εξέτασης: Set the End of the Test. Year, Month, Day, Hour, Minutes

Έτος

2025

Μήνας

8

Ημέρα

11

Ώρα

16

Λεπτό

15

Υποβολή Submission

Tests

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Figure 7. Tutor screen: the test settings. DK-Practice: The Intelligent Educational Recommendations System. Course: Computer Organization and Architecture. Test Archive. New Test, Test's dates.

[illegible]

Figure 8. Tutor screen: overview of the class’s knowledge state. Computer Organization and Architecture. Test date: 2025-01-11, Student name, E-mail, Degree. Temporary accounts had been made to test the Platform’s functionalities.

- with the “*student*” role can choose the course (Figure 9) and perform the tests set by corresponding “*tutor*” (Figure 10). Additionally, each “*student*” has access to previous test performances to view the results. In Figure 11, the results of the user *Student 1* are shown after running a test (for example, the test on 1 May 2025):
 - The success rate in the test, and whether the questions were answered correctly or incorrectly (green color-correct answer, red color-incorrect answer)
 - Personalized recommendations are generated by incorporating educational content and estimating the rate of knowledge mastering through the knowledge tracing component. A lower percentage associated with a given educational content item corresponds to a larger proportion of red in the visual bar representation, indicating knowledge gaps in the concepts associated with this educational content.
 - The knowledge state, represented as the percentage of mastery for each concept, is derived from the knowledge tracing component following the completion of the test. The green segment of the bar chart visualizes the proportion of knowledge acquired in the concepts taught within the specific course.

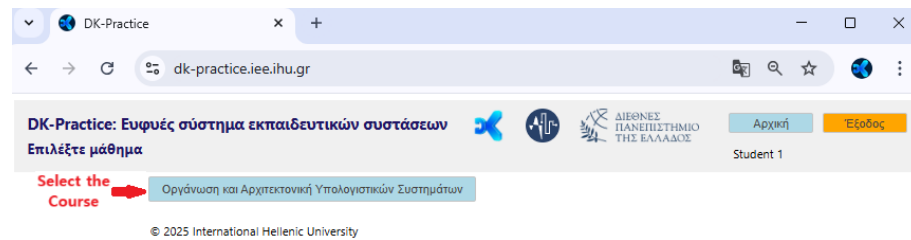


Figure 9. Student screen: selection of the course test. DK-Practice: The Intelligent Educational Recommendations System. Select the course. (Computer Organization and Architecture.)

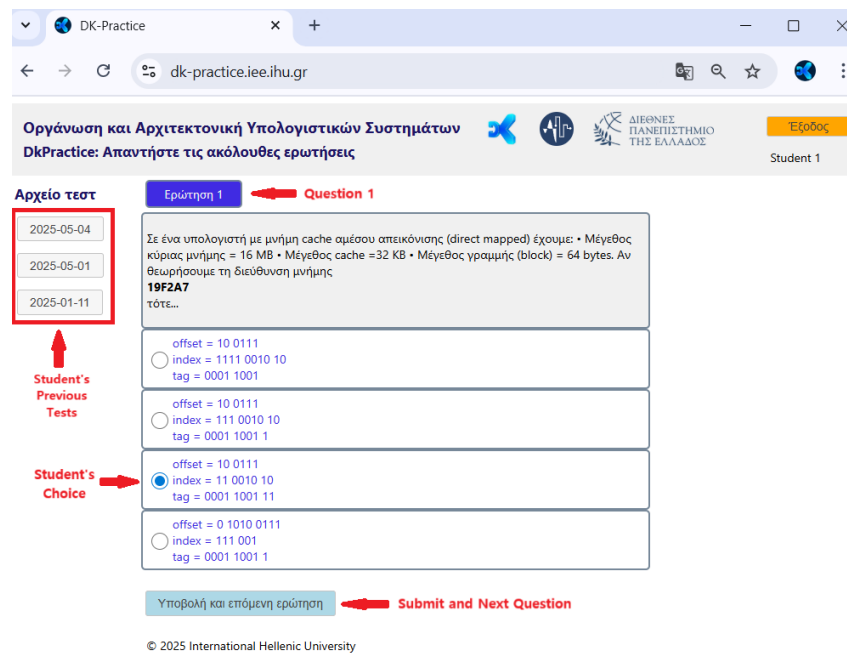


Figure 10. Student screen: answering the test questions. Computer Organization and Architecture. DK-Practice: Answer the questions. Submit the answer and the next question.



Figure 11. Student screen: test results, personalized educational content recommendations, personalized knowledge state according to the learning objects or concepts of the course. Computer Organization and Architecture. The Test date: 1 May 2025. The green color represents knowledge possession, and the red color represents the corresponding knowledge gap.

5. The Case Study

The objective of this case study is to examine and assess the DK-PRACTICE platform and its functionalities within a real learning environment in higher education. To this end, the platform was implemented in the undergraduate course “*Computer Organization and Architecture (COaA)*”, offered in the second year of study within the Department of Information and Electronic Engineering at the International Hellenic University in Greece.

As part of the case study design, two diagnostic assessments were implemented, primarily targeting second-year students enrolled in the course and scheduled to participate in the subsequent official examination period. The first assessment, hereafter referred to as the “**Pre-test**”, was administered approximately one month before the official examination. Its purpose was to evaluate students’ knowledge readiness and to generate personalized recommendations for targeted study based on identified areas of need. Following a two-week interval, a second assessment, hereafter referred to as the “**Post-test**”, was conducted. This assessment was based on the assumption that the students had engaged in the recommended learning resources and had studied the instructional material proposed through the DK-PRACTICE platform.

In both experimental tests, students responded to **10** questions among the set of 120 questions, the outcomes of which were used to assess their knowledge state concerning the 31 concepts covered in the 20 course book [48] chapters (Table 3). To ensure that the assessment aligns with international educational standards, these concepts were mapped to the standardized Body of Knowledge for Computer Science (<https://csed.acm.org/wp-content/uploads/2024/01/Body-of-Knowledge-v1-bookmarksv2.pdf>, accessed on 23 December 2025) as shown in Table 3, where each concept is linked to defined KA and KU, providing a robust curriculum body of knowledge dimension to our data model.

Table 3. Concepts and corresponding COaA course cook chapters according to the Body of Knowledge (BoK) Knowledge Areas (KA) and Knowledge Units (KU). (AR: “Architecture and Organization”; SEP: “Society, Ethics, and the Profession”; SPD: “Specialized Platform Development”).

Concept_Name	Chapter	BOK KA	Knowledge Unit (KU)
CPU structure, Organization and architecture.	1.1	AR	Assembly-Level Machine Organization.
Microelectronics, Moore’s law.	1.2	AR SEP	Digital Logic and Digital Systems. Computing History.
History.	1.3	SEP	Computing History.
Embedded systems, Microcontrollers.	1.5	SPD	Embedded Platforms.
ARM architecture.	1.6	SEP	Computing History.
Cloud computing.	1.7	SEP	Computing History.
Design for performance, Performance balance.	2.1	AR	Performance and Energy Efficiency.
Performance laws.	2.3	AR	Performance and Energy Efficiency.
Execution rate, Processor performance, Performance measures, Order execution rate.	2.4	AR	Performance and Energy Efficiency

Table 3. Cont.

Concept_Name	Chapter	BOK KA	Knowledge Unit (KU)
Interrupts, Instruction cycles, Instruction phases.	3.2	AR	Interfacing and Communication. Assembly Level Machine Organization.
QPI.	3.5	AR	Interfacing and Communication.
PCI.	3.6	AR	Interfacing and Communication.
Cache mapping.	4.3	AR	Memory Hierarchy.
Semiconductor memory, ROM.	5.1	AR	Memory Hierarchy.
Error correction.	5.2	AR	Memory Hierarchy.
DDR memory, SDRAM.	5.3	AR	Memory Hierarchy
Flash memory.	5.4	AR	Memory Hierarchy.
Magnetic disk organization.	6.1	AR	Interfacing and Communication.
RAID.	6.2	AR	Interfacing and Communication.
Programmed I/O.	7.3	AR	Interfacing and Communication.

Based on this assessment, personalized content recommendations were generated to facilitate further study and enhance knowledge acquisition. Apart from the initial question, each subsequent question asked to a student was adaptively determined by the history of previous question responses. This process was supported by the PB-BoW knowledge tracing model, which operates in the backend of the platform.

Student participation in both assessments was strictly voluntary and based on informed consent. Moreover, participation did not influence students' grading outcomes, nor did it interfere with the formal assessment procedures of the official examination period.

5.1. Pre-Test Description

The first, “**Pre-test**”, knowledge tracing test was to examine the knowledge state of the participating students. A total of 138 students participated, all of whom had previously registered and activated their accounts. The test was administered in groups, a procedure determined by the availability of laboratory facilities and computing resources. Two laboratories were used, which contained 25 and 27 internet-enabled workstations, respectively. Throughout testing sessions, the DK-PRACTICE platform demonstrated reliable performance, supporting up to 50 simultaneous users without malfunctions in its services. The tutor monitored the progress of the students in real time through the user interface of the “*tutor*” of the platform (Figure 12).

In addition, the participants were asked to complete a questionnaire to record the user experience regarding the functionality, usability, and environment of the platform. 31 of the 138 participants completed and submitted the questionnaire shared online. Their responses provide valuable insight into the user experience and are further analyzed below in Section 5.3.

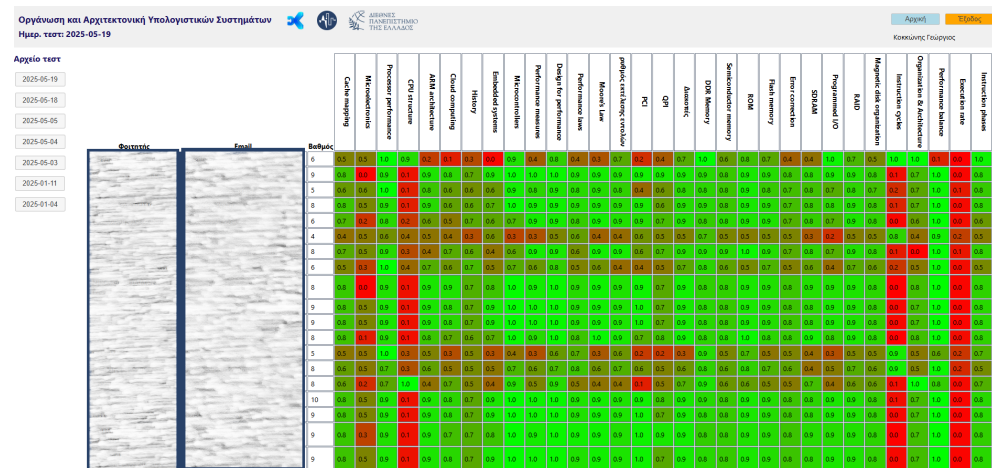


Figure 13. A part of the tutor’s user interface during the “Post-test” process on the DK-PRACTICE platform, showing student outcome monitoring. Computer Organization and Architecture. Test date: 19 May 2025. The knowledge for each student (rows) per concept (columns) is depicted using shades of green and red. The intense red color corresponds to the knowledge gaps, while the intense green corresponds to the possession of knowledge of the concepts.

After the test, participants were asked to complete a questionnaire on the test's effectiveness. A total of 98 students participated in both assessments. Valid responses were received from 68 (69.4%) students who had participated in both assessments and completed the qualitative survey, with the results presented below.

The questions of the second questionnaire, apart from demographic data, are listed by category:

1. Post-Test User Experience

- (a) The presentation of the material was organized and understandable.
- (b) I would use the platform again to test my knowledge.
- (c) My overall experience with the platform has been positive.
- (d) Use of the platform was pleasant.
- (e) I felt that the platform understood my learning needs.
- (f) I feel that the platform has contributed to my progress.
- (g) I have a better picture of my weaknesses after this experience.
- (h) The personalization of the proposals was evident and useful.
- (i) I feel that I am better prepared for the exam.

2. Effectiveness of Recommendations

- (a) The content was relevant to the points I was lacking.
- (b) The suggested material helped me to better understand the subject matter.
- (c) The test of the second experiment recorded my progress.
- (d) The proposals met my real needs.
- (e) My progress from the first to the second test was evident.
- (f) I would use the platform again to test my knowledge.

3. Willingness to Use the Platform by Students

- (a) I would continue to use the platform in other courses.
- (b) I would recommend the platform to other students.

Both questionnaires were created to encompass a wide range of factors influencing student experience, system reliability, quality of personalized recommendations, and perceived learning progress. This multi-layered approach ensures valid and reliable evaluation outcomes, providing evidence-based insights for improving the platform and integrating it effectively into educational practices.

5.3. Evaluation

The evaluation of the platform is conducted according to two primary axes: User Experience and Platform Effectiveness.

1. **User Experience:** In terms of user experience, the evaluation was conducted in two phases. The first phase was implemented during the final stage of system development, during which the platform was subjected to testing to identify usability-related improvements. The evaluation highlighted issues concerning the visual presentation of questions, the display of results upon test completion, and the functionality of the password recovery mechanism. Following the incorporation of the necessary modifications, these issues were resolved, thereby ensuring the platform's readiness for deployment in the "Computer Organization and Architecture" course. The second phase of the usability evaluation was carried out by the students who engaged with the platform during the two testing sessions. In both evaluations, students reported a positive user experience with the DK-PRACTICE platform. The survey results of the "Pre-test", presented in the chart (Figure 14) demonstrate an overall positive evaluation of the intelligent knowledge tracing platform by higher education students. The highest levels of agreement were recorded for the statement "Using the platform was easy and intuitive", with 48.39% agreeing and 38.71% strongly agreeing, indicating that the platform's usability is one of its strongest features. Similarly, the layout of the environment was well received, with 41.94% agreeing and 38.71% strongly agreeing that it was logical and clear.

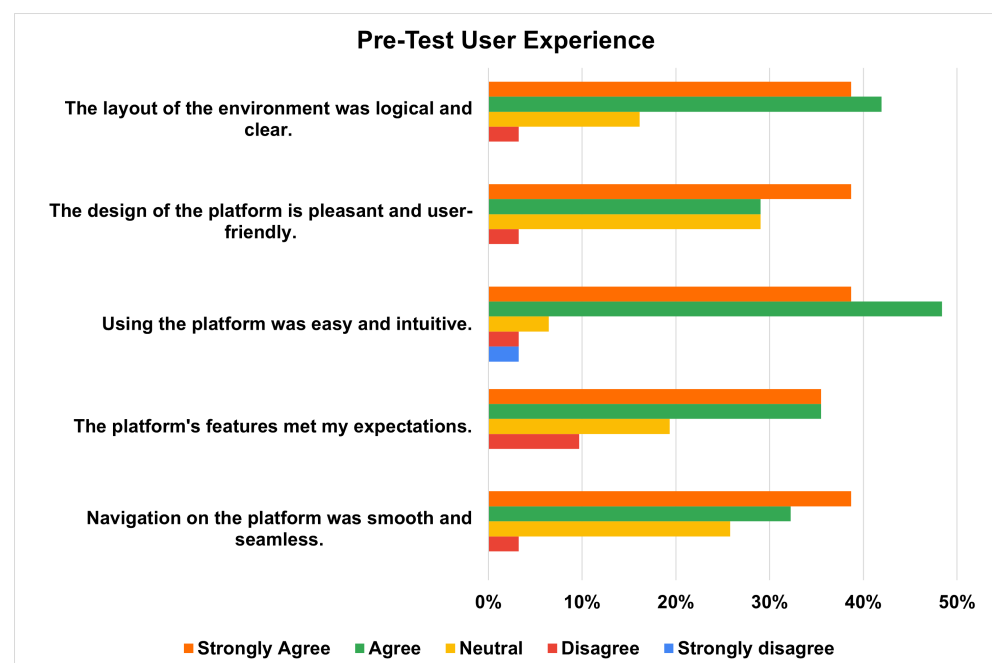


Figure 14. Pre-test: user experience evaluation of using DK-PRACTICE platform by students.

The design of the platform was also evaluated positively, with 29.03% agreement and 38.71% strong agreement, though this item received the highest proportion of neutral responses (29.03%). Regarding functionality, 35.48% of students agreed and another 35.48% strongly agreed that the platform's features met their expectations, although 9.68% disagreed, suggesting some room for improvement. Navigation was perceived as smooth and seamless by most respondents (32.26% agreed, 38.71% strongly agreed), although 25.81% remained neutral. In summary, the findings show that the DK-PRACTICE platform is largely intuitive, user-friendly, and effective, with only a small minority reporting dissatisfaction. These results highlight strengths in

ease of use and logical design, while pointing to opportunities for further refinement of features and overall design appeal.

Based on questionnaire answers after the “**Post-test**” (Figure 15), most respondents expressed satisfaction, with notable percentages agreeing or strongly agreeing with key statements about platform usefulness. Specifically, 72.06% felt the material was well-organized, understandable, and enjoyable to use. Additionally, students overwhelmingly believed the platform was beneficial for their learning; a combined 73.53% of users agreed or strongly agreed that it contributed to their progress and helped them identify their weaknesses more effectively. The most positive feedback centered on the intention to reuse the platform, as 70.59% of respondents would use it again to test their knowledge, demonstrating strong support for its effectiveness as a learning tool.

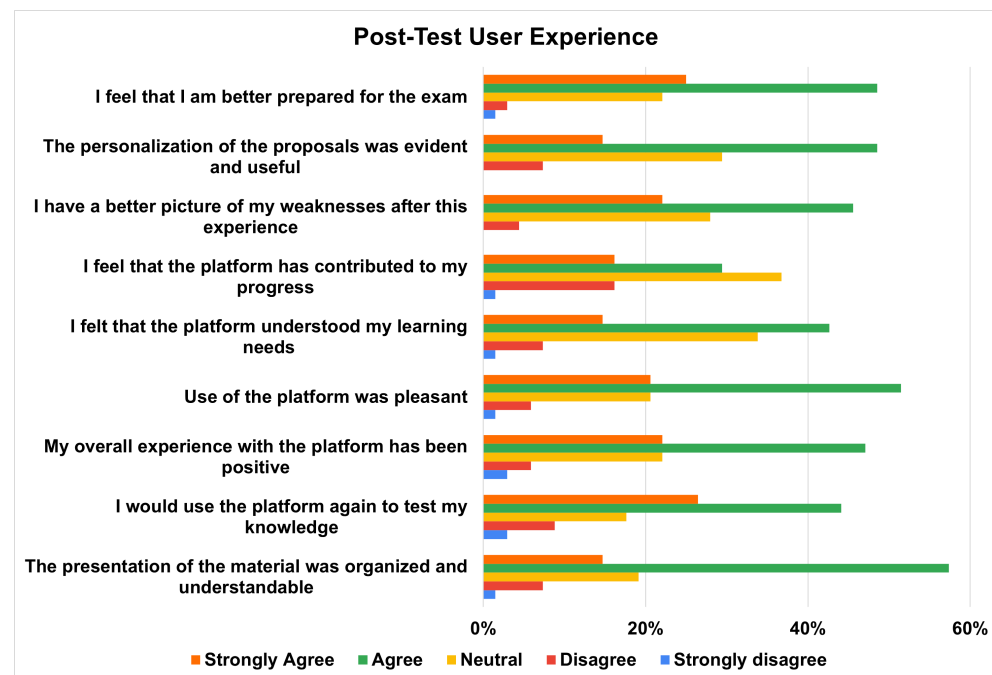


Figure 15. Post-test: user experience evaluation of using DK-PRACTICE platform by students.

2. **Platform Effectiveness:** This axis relates to the effectiveness of the services offered, specifically the assessment of knowledge state and the educational recommendations during pre-/post-tests performed by students.

The findings of the quality assurance questionnaire after “**Pre-test**” indicate that students mostly viewed the intelligent knowledge tracing platform DK-PRACTICE primarily positively (Figure 16). Most respondents chose “*Agree*” on all five items, with especially high agreement for the statements “*This experience helped me understand where I needed improvement*” and “*I trust the knowledge assessment process followed by the platform*”, both surpassing 60%. A significant number of students also strongly agreed, especially regarding the platform’s ability to identify individual strengths and weaknesses and accurately assess their knowledge. Neutral responses were moderate, while reports of disagreement or strong disagreement were minimal. These results show that students generally see the platform as effective, reliable, and helpful for their learning, highlighting its potential value in higher education.

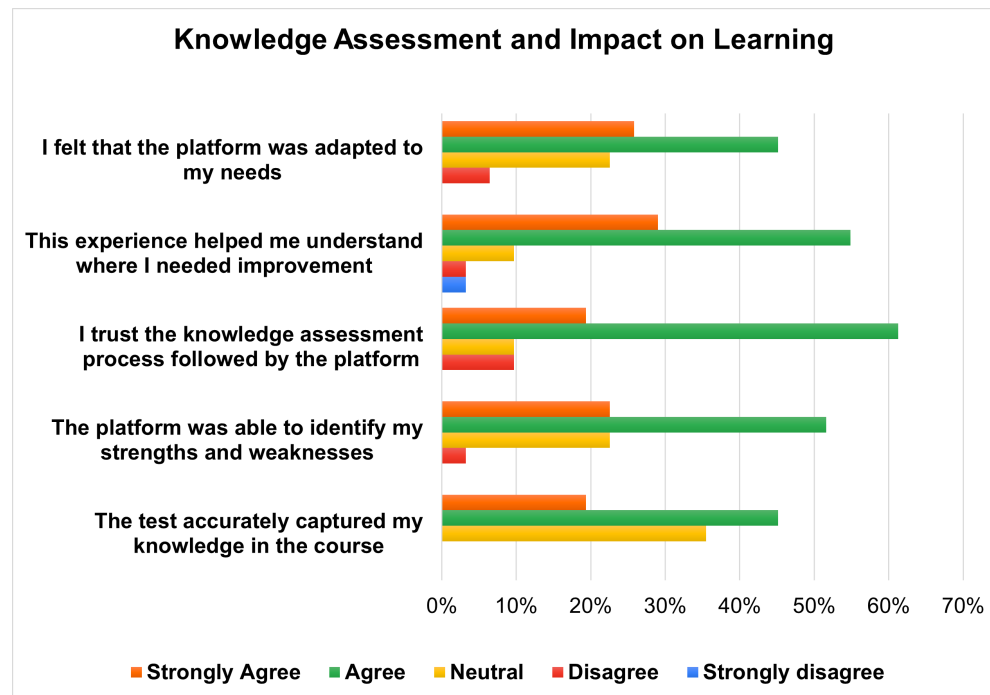


Figure 16. Pre-test: knowledge assessment and impact on learning through the DK-PRACTICE platform.

The “**Post-test**” quality assurance questionnaire, given to higher education students, highlights both the high use and perceived effectiveness of the DK-PRACTICE platform’s recommendation system. As shown in Figure 17, most respondents actively engaged with the recommendations. Specifically, 36.76% reported a moderate level of use (score of 3 on a 5-point scale), while a significant number of students indicated a high degree of use, with 30.88% selecting a score of 4 and 16.18% choosing the highest score of 5. These results suggest that the platform’s recommendations were considered valuable enough to encourage active engagement, supporting the platform’s role as a tool for personalized learning.

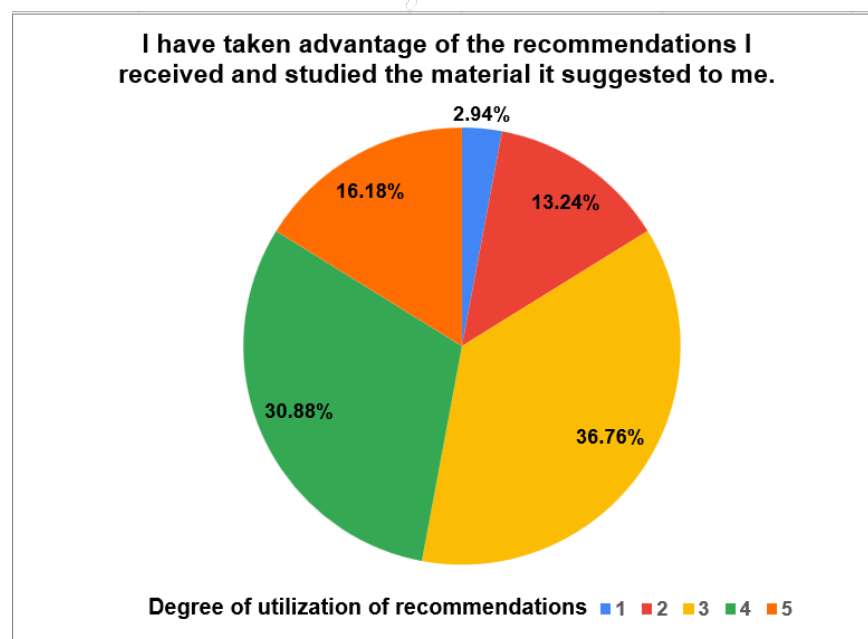


Figure 17. Post-test: the degree of utilization of recommendations.

Further analysis of the survey data, as shown in Figure 18, confirms the high effectiveness of these recommendations from the students' point of view. Most students agreed or strongly agreed with the statements, indicating a positive user experience. Notably, over 50% of respondents agreed that *"The content was relevant to the points I was lacking"* and that *"The proposals met my real needs"*. Additionally, a large percentage of students felt that the *"suggested material helped me to understand the subject matter better"* and that *"The test of the second experiment recorded my progress"*.

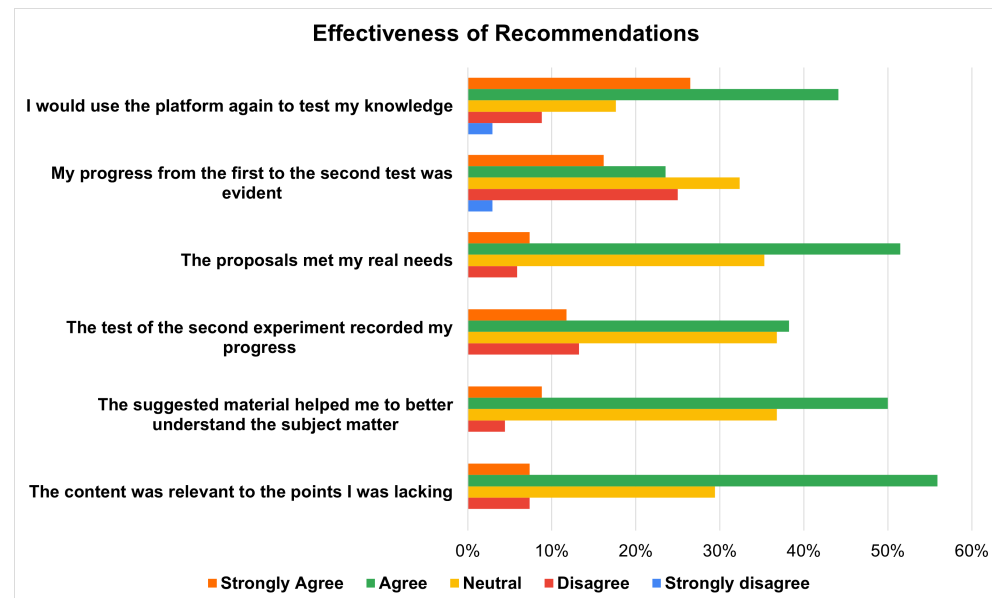


Figure 18. Post-test: effectiveness of recommendations.

These results collectively highlight the platform's success in providing targeted and useful content that genuinely helps students improve academically.

The findings of the quality assurance questionnaire after both tests show a strong willingness among students to continue using the DK-PRACTICE] platform and to recommend it to others.

As illustrated in the Figure 19, most respondents chose higher ratings (4 and 5 on a five-point scale), with 40% indicating that they would continue using the platform in other courses and 36% stating that they would recommend it to fellow students. Furthermore, 22.58% of students reported the highest level of willingness (5) to continue using the platform, while 27.1% showed the highest intention to recommend it. Lower ratings (1 and 2) were minimal, indicating very little resistance to future adoption or endorsement of the system. In general, the students valued its usability, design, and potential as a learning tool.

Despite positive feedback, students identified areas for improvement. In **"Pre-test"**, technical issues such as system crashes were noted, but local network problems caused these. In **"Post-test"**, participants recommended features such as a timer, question navigation, a skip option, and semester access for continuing practice. Overall, the findings indicate that the platform effectively fulfills student learning needs and is seen as a valuable and efficient educational resource.

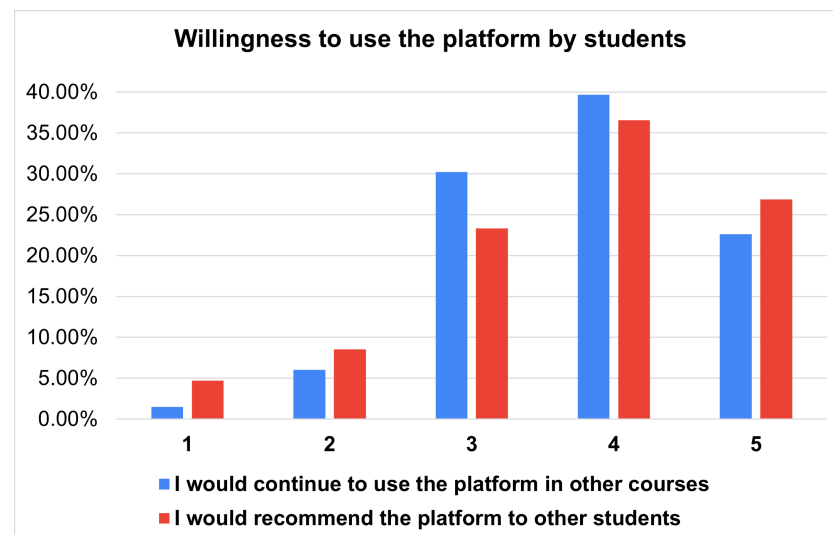


Figure 19. Willingness to use the platform DK-PRACTICE platform by students.

6. Results and Discussion

The deployment of DK-PRACTICE illustrates the practical potential of integrating machine learning-based knowledge tracing with personalized recommendation mechanisms in higher education. By accurately identifying individual learning gaps and providing tailored instructional content, the platform supports both academic achievement and self-regulated learning. Its lightweight yet effective PB-BoW model not only outperforms deeper architectures in predictive accuracy but also offers efficiency in training and scalability for large classrooms. The evaluation of DK-PRACTICE focused on two interrelated dimensions: the technical performance of its knowledge tracing model and the platform's educational effectiveness in a real classroom environment. The predictive capacity of the proposed PB-BoW KT model was benchmarked against the KT-BiGRU, SAKT, and DKT models using three datasets: "ASSISTment 2009", "ASSISTment 2017", and the newly constructed "COaA Course" dataset. In all cases, the PB-BoW approach consistently outperformed all baseline models across both AUC and accuracy metrics, establishing its superior predictive power. On the "ASSISTment 2009" dataset, PB-BoW recorded an AUC of 0.765, notably exceeding the 0.723 achieved by both SAKT and DKT. Similarly, in the "ASSISTment 2009" dataset, PB-BoW maintained its lead with an AUC of 0.733, while the SAKT and DKT models trailed significantly with scores of 0.650 and 0.644, respectively. The largest performance gap appeared in the "COaA" dataset, where PB-BoW achieved an AUC of 0.805 and an accuracy of 0.787, substantially outperforming the 0.770 (AUC) of the transformer-based SAKT and the 0.756 (AUC) of the DKT model.

These results suggest that the PB-BoW model provides more robust predictions, generalizes effectively across datasets of varying size and complexity, and does so with considerably lower computational requirements. Training was conducted efficiently on a standard CPU without the need for GPUs, requiring only around five seconds per epoch, which underscores the model's suitability for practical classroom applications where resources may be limited.

To validate the performance and effectiveness of the model, we examined its statistical significance compared to the other models. Table 4 shows that PB-BoW achieved the highest accuracy (0.787), outperforming DKT (0.738), SAKT (0.739), and KT-BiGRU (0.719) by a notable margin. Statistical validation further confirms these findings, as the ANOVA test yielded a p -value < 0.001 , indicating a highly significant difference across all models. Moreover, post-hoc paired t -tests using bootstrap resampling revealed that the performance gain of PB-BoW over DKT, SAKT, and KT-BiGRU is statistically significant in every case.

(p -value < 0.001). The exceptionally high t -statistics reported in Table 4 (DKT: 89.525, SAKT: 63.940, KT-BiGRU: 110.810) serve to underscore the clear superiority of our proposed approach. The robustness of the PB-BoW model is also reflected in its 95% confidence interval for AUC [0.786, 0.825], which consistently remains above the mean AUC of the competing models. These results suggest that the BoW-inspired representation effectively captures the essential features of student interactions within the “COaA” context, leading to more accurate knowledge tracing predictions.

Table 4. Performance comparison of PB-BoW against DKT, SAKT, and KT-BiGRU on the “COaA” dataset. Statistical significance was assessed via ANOVA and post-hoc bootstrap t -tests (p -value < 0.001 for all comparisons).

Model	Accuracy	Mean AUC	p -Value	t -Statistic
PB-BoW	0.787	0.805	-	-
DKT	0.738	0.756	<0.001	89.525
SAKT	0.739	0.770	<0.001	63.940
KT-BiGRU	0.719	0.754	<0.001	110.810

The consistent superiority of the PB-BoW model over DKT, SAKT, and KT-BiGRU architectures across three different datasets (“COaA”, “ASSIST2009”, “ASSIST2017”) highlights a crucial theoretical dimension: the resilience of statistical normalization over structural complexity. While deep learning models attempt to simulate learning dynamics through complex recursive portals and attention mechanisms, they often become vulnerable to the noise of educational data, leading to overfitting. The success of PB-BoW is due to the simplicity of the Bag-of-Words representation which acts as a regulatory factor that eliminates unnecessary parameters, focusing on the student’s cumulative knowledge state. Instead of being consumed in the fragile modeling of the exact time sequence, which is often disrupted by extrinsic factors in educational practice, PB-BoW condenses the essence of knowledge into fixed and compact features, demonstrating that information density is often preferable to architectural depth.

The case study in the “Computer Organization and Architecture” course provided an opportunity to evaluate how effectively the platform identifies knowledge gaps and supports learning progress. A total of 138 students participated in the “Pre-test” phase, which aimed to establish baseline knowledge levels across 31 concepts. The results of the “Pre-test” confirmed that the system could identify strengths and weaknesses with a high degree of accuracy. Survey responses indicated that more than sixty percent of students trusted the system’s ability to reflect their actual knowledge state, and a majority also reported that the assessment helped them to recognize areas in which improvement was necessary. These findings highlight the role of the platform not only as a diagnostic tool but also as a means of raising students’ awareness of their own learning needs.

As shown in Figure 20, the results of the post-test are consistently higher than those of the pre-test across most course concepts. Gains are particularly pronounced in topics such as “QPI”, “DDR Memory”, “Interrupts”, and “Magnetic disk organization”, where the post-test results clearly exceeded the pre-test baselines. This improvement demonstrates that the system’s recommendations were not only relevant but also effective in guiding students toward addressing their weaker areas. Some concepts, such as “Execution rate” and “Instruction phases”, showed relatively smaller improvements, suggesting areas where further refinement of the recommendation mechanism or expansion of content resources may be required. Overall, the comparative analysis indicates that students benefited significantly from the adaptive learning process, validating the platform’s ability to translate diagnostic assessments into tangible learning outcomes.

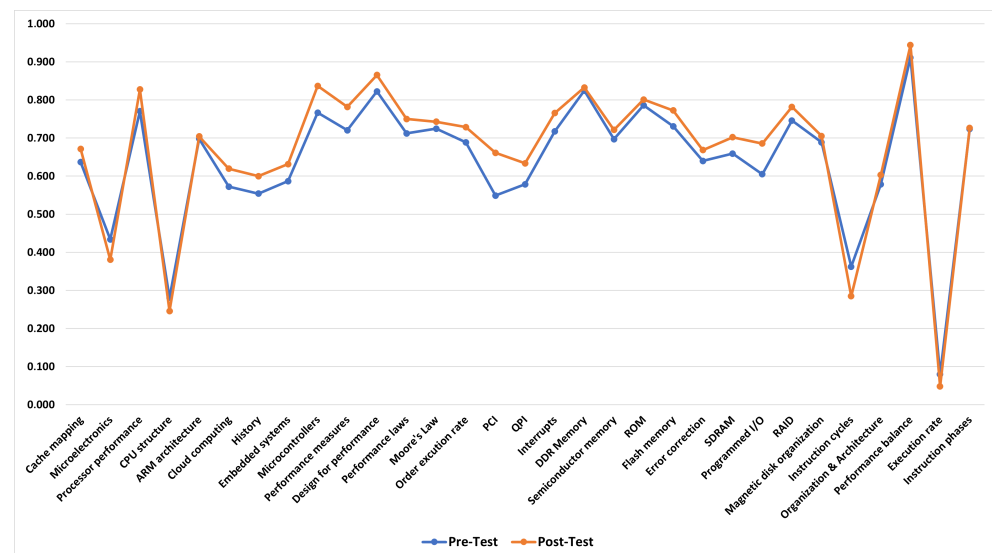


Figure 20. Comparison of “Pre-test” (blue line) and “Post-test” (orange line) performance across 31 course concepts. The results illustrate consistent improvement in student knowledge following engagement with the personalized recommendations provided by DK-PRACTICE.

The results from student questionnaires further support the effectiveness of the platform. During the “Pre-test”, more than 60% of respondents expressed trust in the system’s ability to accurately assess their knowledge and identify strengths and weaknesses. Many students reported that the test helped them gain a clearer understanding of where improvement was needed. In the “Post-test” survey, approximately three-quarters of the participants stated that the platform contributed positively to their progress, with most agreeing that the recommendations were both relevant and useful. Additionally, a large majority expressed willingness to continue using the platform in other courses and to recommend it to peers.

Taken together, these findings highlight the dual strengths of DK-PRACTICE: its robust predictive accuracy and its capacity to generate meaningful educational improvements. The chart comparing Pre- and Post-test results provides direct evidence that the system not only diagnoses knowledge gaps but also helps students to address them, leading to measurable learning gains within a short timeframe. Importantly, the PB-BoW model’s simplicity and computational efficiency show that advanced deep learning architectures are not always necessary to achieve strong results in knowledge tracing, especially when the goal is practical deployment in educational contexts.

The students’ positive evaluations further reinforce the value of integrating knowledge tracing with personalized recommendations. The willingness of students to continue using the platform in future courses indicates both satisfaction with its current design and confidence in its potential for broader application. At the same time, the limited improvements in certain topics reveal opportunities for refining the recommendation algorithms and enriching the content base to ensure consistent benefits across all course areas.

In summary, the results demonstrate that DK-PRACTICE successfully bridges the gap between AI-based research and classroom practice. By providing accurate assessments, effective recommendations, and measurable learning gains, the platform emerges as a scalable and practical tool for advancing personalized learning in higher education.

7. Conclusions and Future Works

This study presented DK-PRACTICE, an intelligent platform that integrates knowledge tracing with personalized recommendations to enhance learning in higher education. The proposed PB-BoW model consistently outperformed a recurrent neural network

baseline, achieving higher AUC and accuracy across benchmark datasets and the newly developed “COaA Course” dataset. Its efficiency and low computational requirements demonstrate that robust predictive performance can be achieved without complex deep learning architectures, making the model practical for real classroom deployment.

The case study in the “Computer Organization and Architecture” course confirmed the platform’s educational impact. **Pre-** and **Post-test** comparisons revealed measurable improvements in student knowledge in most of the concepts evaluated, while surveys showed that students trusted the assessments of the system, valued the relevance of its recommendations, and expressed a strong willingness to reuse them in future courses. These findings validate DK-PRACTICE as both technically reliable and pedagogically effective, successfully bridging the gap between AI-based research and classroom practice. Beyond technical reliability, the integration of the curriculum body of knowledge into the platform’s data model ensures that identified knowledge gaps are directly mapped to standardized educational goals, making the system a valuable asset for formal curriculum assessment. Furthermore, this alignment with practical educational needs underscores the platform’s ability to facilitate competence-based learning by shifting the focus from simple content delivery to the verified mastery of specific course competencies.

The findings of this study provide comprehensive answers to the research questions (RQs) posed at the outset. Regarding RQ1, the PB-BoW model demonstrates that a lightweight, non-recurrent approach can achieve high predictive accuracy in knowledge tracing. By decoupling the current question from the knowledge state through a bipolar representation of student–question interactions, the model achieved an AUC of 0.805 and an accuracy of 0.787 on the course dataset. Crucially, the model’s computational efficiency—requiring only 5 s per epoch on a standard CPU—proves that effective KT can be implemented without the need for expensive GPU infrastructure, making it highly suitable for real-time educational environments. In relation to RQ2, the impact of personalized content recommendations was clearly reflected in the improved student learning outcomes. The comparative analysis between **Pre-test** and **Post-test** results showed an increase in student performance across most course concepts. The pedagogical effectiveness is further supported by student feedback, with 73.53% of participants acknowledging that the platform facilitated their progress and over 50% confirming the high relevance of the recommended material to their specific knowledge gaps. By targeting these gaps, DK-PRACTICE operationalizes the core tenets of CBL, ensuring that student progress is predicated on the attainment of proficiency in individual skills rather than a uniform instructional pace.

Concerning RQ3, the implementation of DK-PRACTICE proves that the platform can substantially enhance the educational process by providing self-regulated learning with a reliable diagnostic tool. The system’s robustness was verified in a real-world setting, successfully supporting 50 concurrent users during laboratory sessions. Given its scalability, low computational overhead, and the high level of student acceptance (with 70.59% expressing intent to reuse the platform), DK-PRACTICE stands as a viable and extensible solution for various academic disciplines and institutions seeking to integrate AI-driven personalization into their curricula.

The present research demonstrates that architectural complexity is not a prerequisite for achieving high accuracy in knowledge tracing. The PB-BoW model demonstrated that an approach based on the systematic summation of interactions can achieve superior generalization with statistically significant results (p -value < 0.001). The findings highlight that in data with intrinsic noise, the stability of feature representation outweighs the volatile hidden states that characterize more complex neural networks. The effectiveness of PB-BoW adds new foundations for the development of computationally efficient systems,

demonstrating that a strategic focus on essential information offers reliable predictions and can overcome architecturally complex neural network models.

Despite the encouraging results and the technical robustness of the DK-PRACTICE platform, certain limitations must be acknowledged. First, the scope of the empirical evaluation was confined to a single academic institution and a specific course on computer organization and architecture. While this case study provides solid proof of concept, the generalizability of our findings across diverse academic disciplines and more heterogeneous student populations remains to be fully established through broader deployment.

Furthermore, the current implementation relied on a finite database of educational resources. Although the recommendation engine effectively identified knowledge gaps, the impact of the interventions could be further amplified by integrating a more extensive and multi-modal repository of learning materials, such as interactive simulations and video tutorials, to cater to different learning styles. Finally, the assessment of learning effectiveness was primarily based on immediate gains observed between **Pre-** and **Post-tests**. Consequently, the long-term retention of knowledge and the platform's influence on students' cumulative academic trajectory have not yet been evaluated through longitudinal data. Addressing these constraints constitutes a key priority for the next stages of our research.

To build upon this work, future research will focus on several key directions. First, enhancing the recommendation engine and expanding the content database will ensure more consistent improvements across all concepts. Second, the integration of advanced models, such as attention-based or hybrid approaches, may further improve predictive accuracy while preserving efficiency. Third, scaling the platform across multiple disciplines and institutions, complemented by longitudinal studies, will provide evidence of its broader applicability and long-term educational benefits. Student feedback also points to practical improvements, such as timers, navigation features, and extended access, that could improve usability and engagement.

In addition, we intend to expand the platform for use in future years' courses to include the skills acquired during studies. By extension, the overall picture of knowledge development could be included, with the ultimate goal of analyzing needs and filling curriculum gaps by aligning the data model with the standardized CBoK. This will allow for adapting the curriculum within a comprehensive CBL framework, as well as expanding on the development of a multi-skill model and the integration of LLM/SLM models in the production of educational recommendations.

In conclusion, DK-PRACTICE underscores the transformative potential of integrating robust knowledge tracing with intelligent recommendation systems to create a truly adaptive learning experience. By successfully bridging technical innovation with practical classroom application, this study offers a scalable and effective blueprint for personalized education. With further refinement and broader institutional adoption, DK-PRACTICE stands as a powerful catalyst for evolving higher education into a more data-informed, student-centered, and personalized environment, ultimately empowering both educators and learners to achieve their full potential.

Author Contributions: Conceptualization, M.D. and K.D.; investigation, M.D.; software, M.D.; methodology, M.D. and K.D.; resources, M.D., K.D., and A.S.; data curation, M.D.; validation, M.D., K.D., and G.K.; writing—original draft preparation, M.D.; supervision, K.D.; writing—review and editing, K.D., A.S., G.K., G.E., and D.K. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Ethical review and approval were waived for this study due to it being conducted within the context of a course.

Informed Consent Statement: Informed consent was obtained from all subjects involved in the study.

Data Availability Statement: The GitHub link will be available upon request.

Conflicts of Interest: The authors declare no conflicts of interest.

Abbreviations

The following abbreviations are used in this manuscript:

AI	Artificial Intelligence
BKT	Bayesian Knowledge Tracing
BoK	Body of Knowledge
BoW	Bag of Words
CBoK	Curriculum Body of Knowledge
DKT	Deep Knowledge Tracing
DKVMN	Key-Value Memory Networks
EKT	Exercise-Enhanced Sequential Modeling
GIKT	Graph-based Knowledge Tracing
GPU	Graphics Processing Unit
IRT	Item Response Theory
KA	Knowledge Area
KSGAN	Knowledge Structure-aware Graph Attention Network
KT	Knowledge Tracing
KT-BiGRU	Knowledge Tracing Bidirectional Gated Recurrent Unit
KU	Knowledge Unit
ML	Machine Learning
PB-BoW	Paired-Bipolar Bag-of-Words
RNN	Recurrent Neural Network
SAKT	Self-Attentive Knowledge Tracing

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